电子设计基础训练

电子信息工程学院
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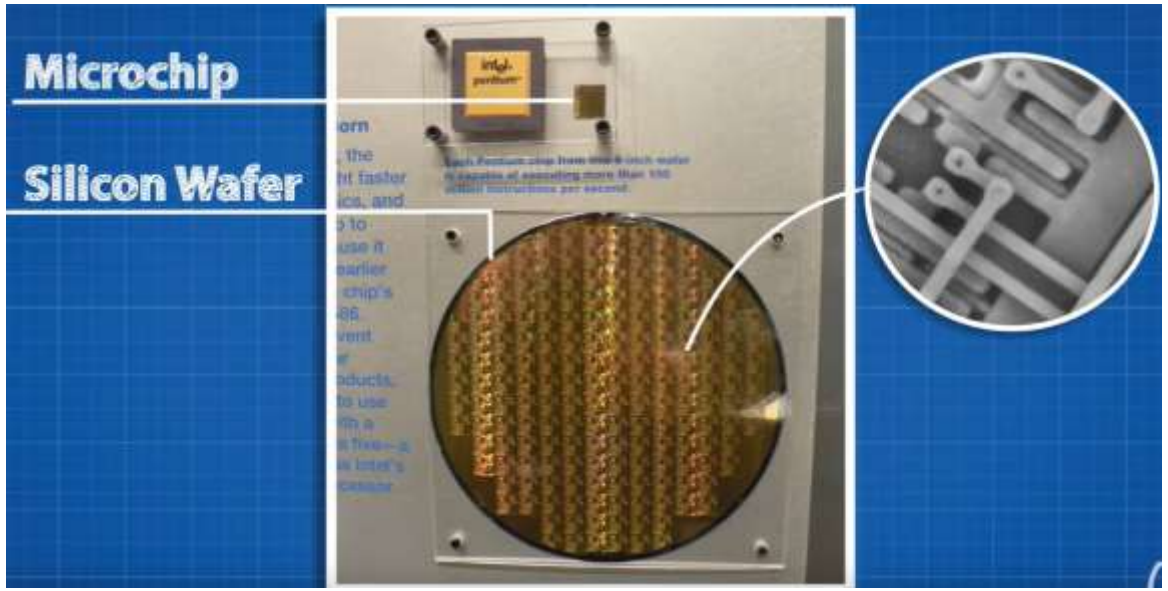
电子设计基础训练

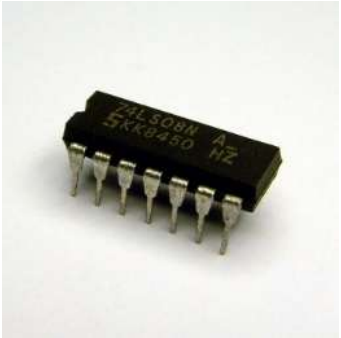
- 逻辑概念与逻辑电平
- 逻辑运算
- 组合逻辑与时序逻辑
- 二极管、三极管搭建门电路
- 常用逻辑门芯片认识与测量
- 译码器、比较器、选择器芯片认识与测量
- 555振荡器认识与测量
- 触发器芯片认识与测量
- 计数器与移位寄存器芯片认识与测量

逻辑概念与逻辑电平

■ 数字逻辑电路定义

- 定义：一种基于二进制数字逻辑设计和运行的，用于处理数字信号的电子电路，也称为数字电路

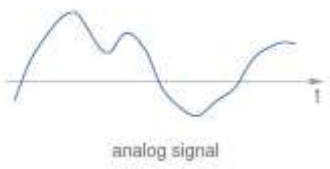




逻辑概念与逻辑电平

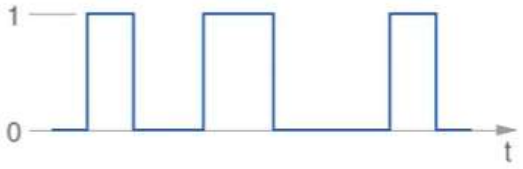
■模拟信号、电路

- 信号形式：在时间上和数值上均是连续的
- 表示方式：模拟自然信号、用连续的电压信号表示信息
- 分析方法：如何不失真的信号传输、处理



■数字信号、电路

- 信号形式：在时间上和数值上均是离散的
- 表示方式：采样、量化用离散的电压序列（0/1编码）表示信息
- 分析方法:输入和输出信号的逻辑关系

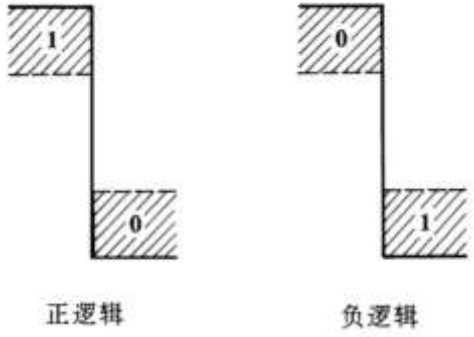


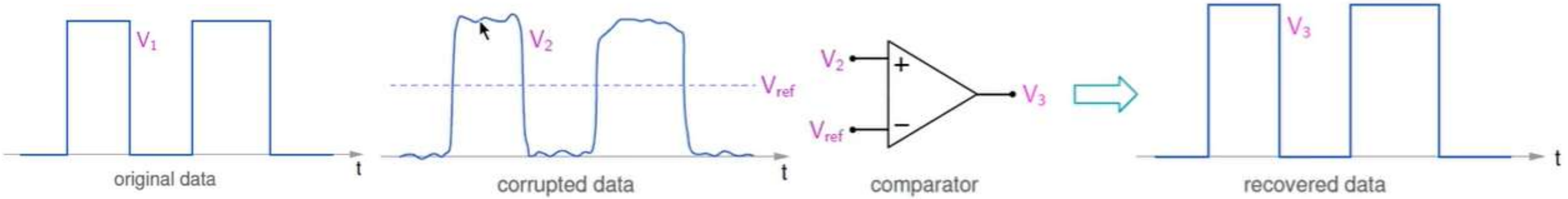
逻辑概念与逻辑电平

■数字电路

- ➢数字电路用电压的高/低电平表示逻辑状态1/0的。此处的高/低电压不是指的某个特定的值，而是一定范围的电平。

- ➢有一定的容错能力，系统可靠性强。





逻辑概念与逻辑电平

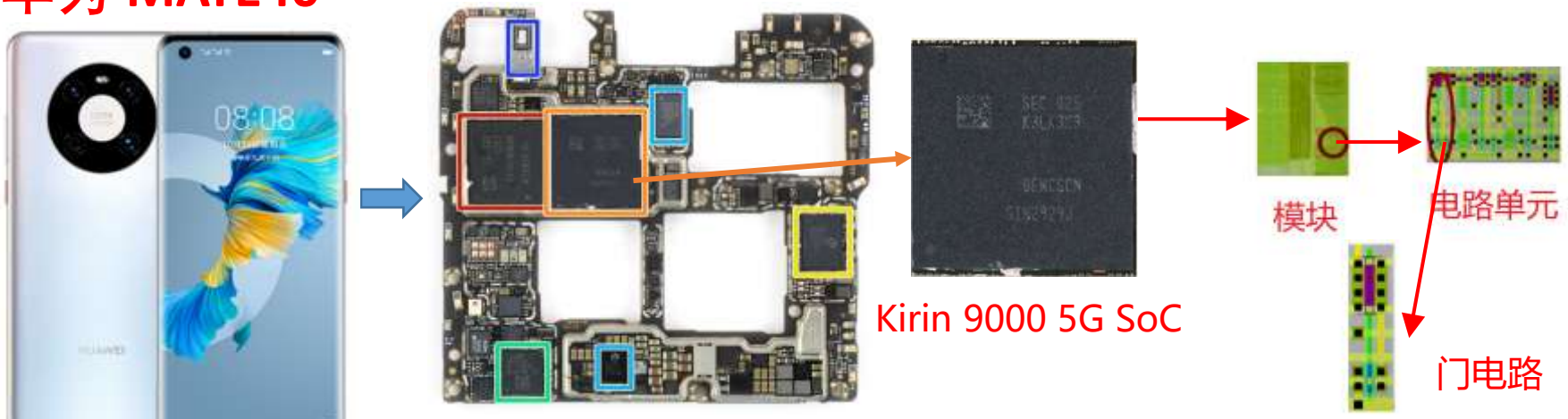
■常用逻辑高低电平

- ➢5V CMOS输入大于3.5V算高电平，输入小于1.5V算低电平
- ➢RS232采用的是负逻辑：逻辑“1”：-5V至-15V，逻辑“0”：+5V至+15V
- ➢5V TTL输入大于2V算高电平，输入小于0.8V算低电平，本次实验所用芯片为74LS系列，均采用该标准

数字系统组成

- 复杂的数字系统都是由最基础的逻辑门电路按一定规则有层次的组成。

华为 MATE40

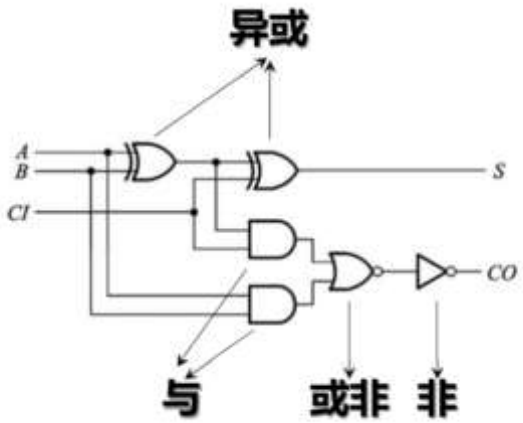


- 有层次的设计即从单一元件到系统，从简单到复杂，逐级设计。符合人类的逻辑思路，也更容易理解。

基本门电路及布尔代数

■ 门电路

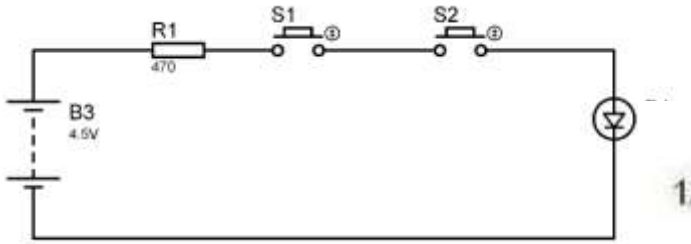
- 定义：实现基本逻辑运算、复合逻辑运算的单元电路，如与门、或门、非门等。
- 逻辑运算：布尔代数（也称为逻辑代数）中定义的与、与非、或非逻辑运算。

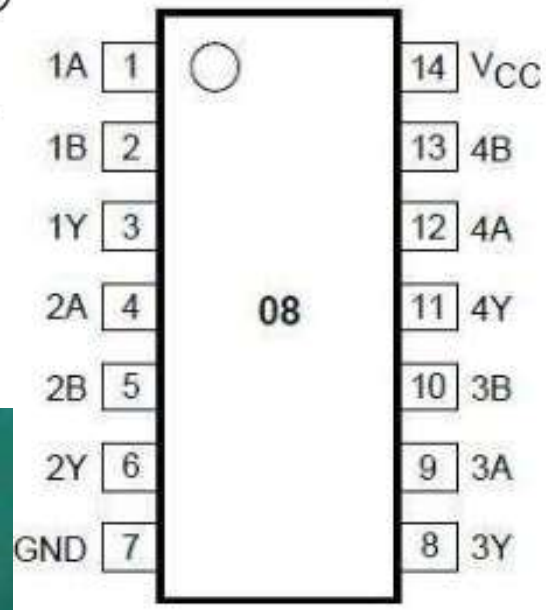


基本逻辑运算

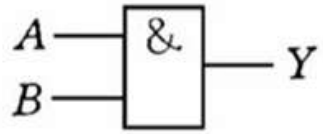
■ 与

- ➢ 条件同时具备，结果发生
- ➢ Y=A AND B = A&B = A·B = AB





A	B	Y
0	0	0
0	1	0
1	1	1
1	0	0

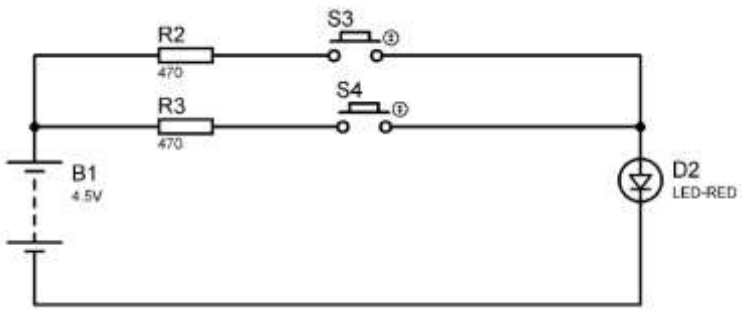


国家标准

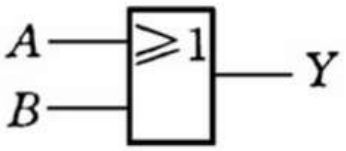
国际标准

基本逻辑运算

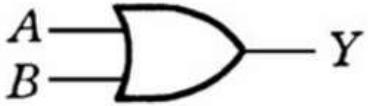
- 或
  - 条件之一具备，结果发生
  - Y=A OR B = A+B

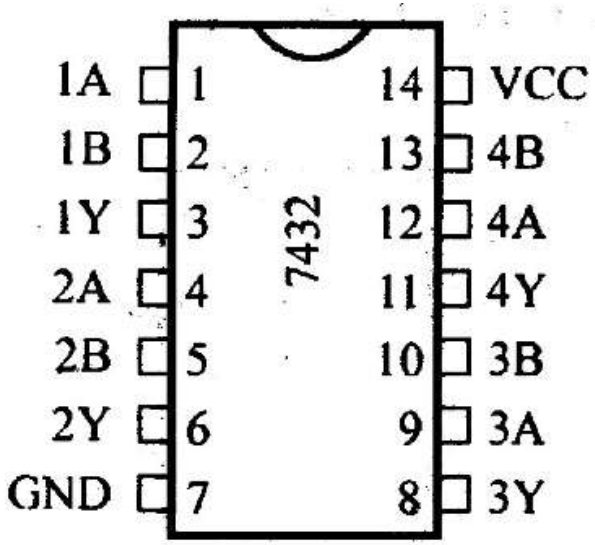


A	B	Y
0	0	0
0	1	1
1	1	1
1	0	1



或

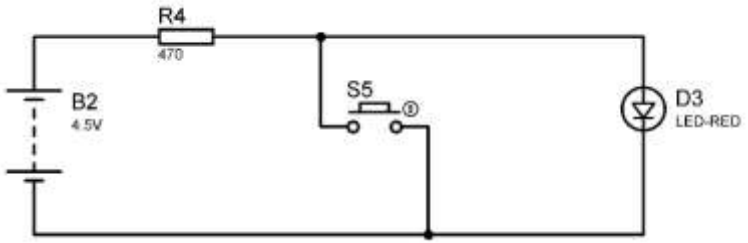




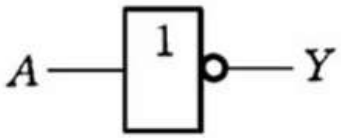
基本逻辑运算

■ 非

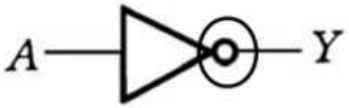
- ➢ 条件不具备，结果发生
- ➢ $Y = NOTA = \bar{A} = A'$

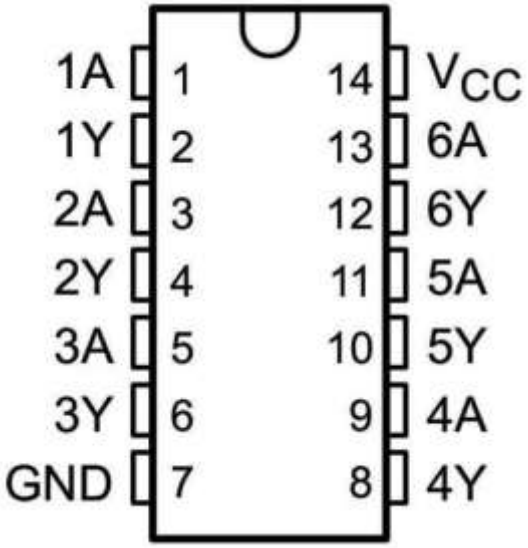


A	Y
0	1
1	0



非



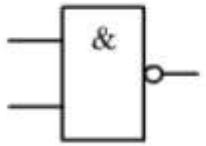


复合逻辑运算

■ 与非

- 条件同时具备，结果不发生
- $Y = \overline{A \cdot B}$

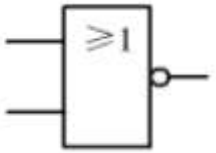
A	B	Y
0	0	1
0	1	1
1	1	0
1	0	1

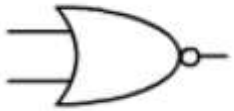


■ 或非

- 条件同时不具备，结果发生
- $Y = \overline{A + B}$

A	B	Y
0	0	1
0	1	0
1	1	0
1	0	0



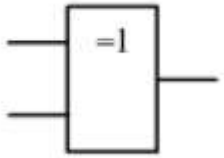


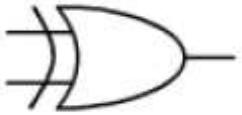
复合逻辑运算

■ 异或

- 条件不一致，结果发生
- $Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	1	0
1	0	1

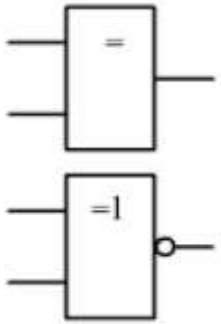


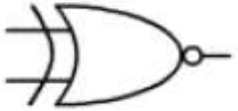


■ 同或

- 条件一致，结果发生
- $Y = \overline{A \oplus B}$

A	B	Y
0	0	1
0	1	0
1	1	1
1	0	0





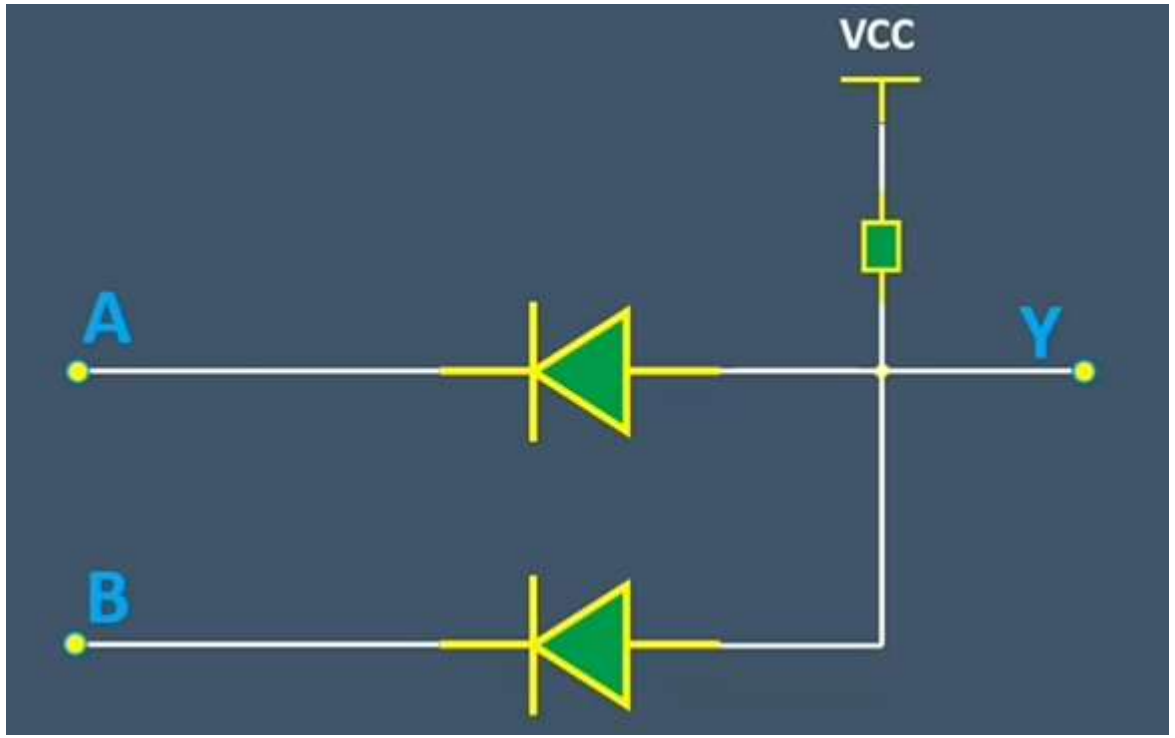
组合逻辑与时序逻辑

- 组合逻辑
  - 无记忆
  - 电路当前的输出完全由当前的输入决定，与之前的输入无关或者说与当前的状态无关
  - 典型的组合逻辑电路
    - 加法器，减法器，多路器等

- 时序逻辑
  - 有记忆
  - 输出不仅与当前时刻输入变量的取值有关而且与电路的原状态有关
  - 典型的时序逻辑电路
    - 锁存器、触发器、寄存器、移位寄存器、计数器等

二极管设计之两输入与门电路--直流输入

■ 与门电路图、真值表

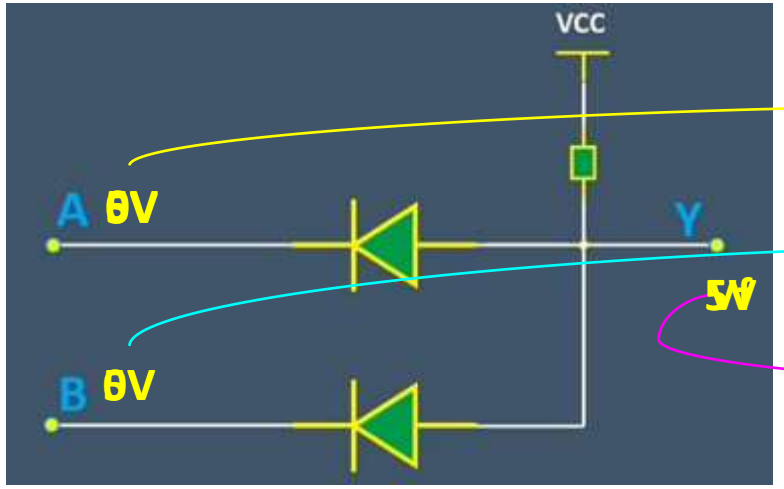


A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

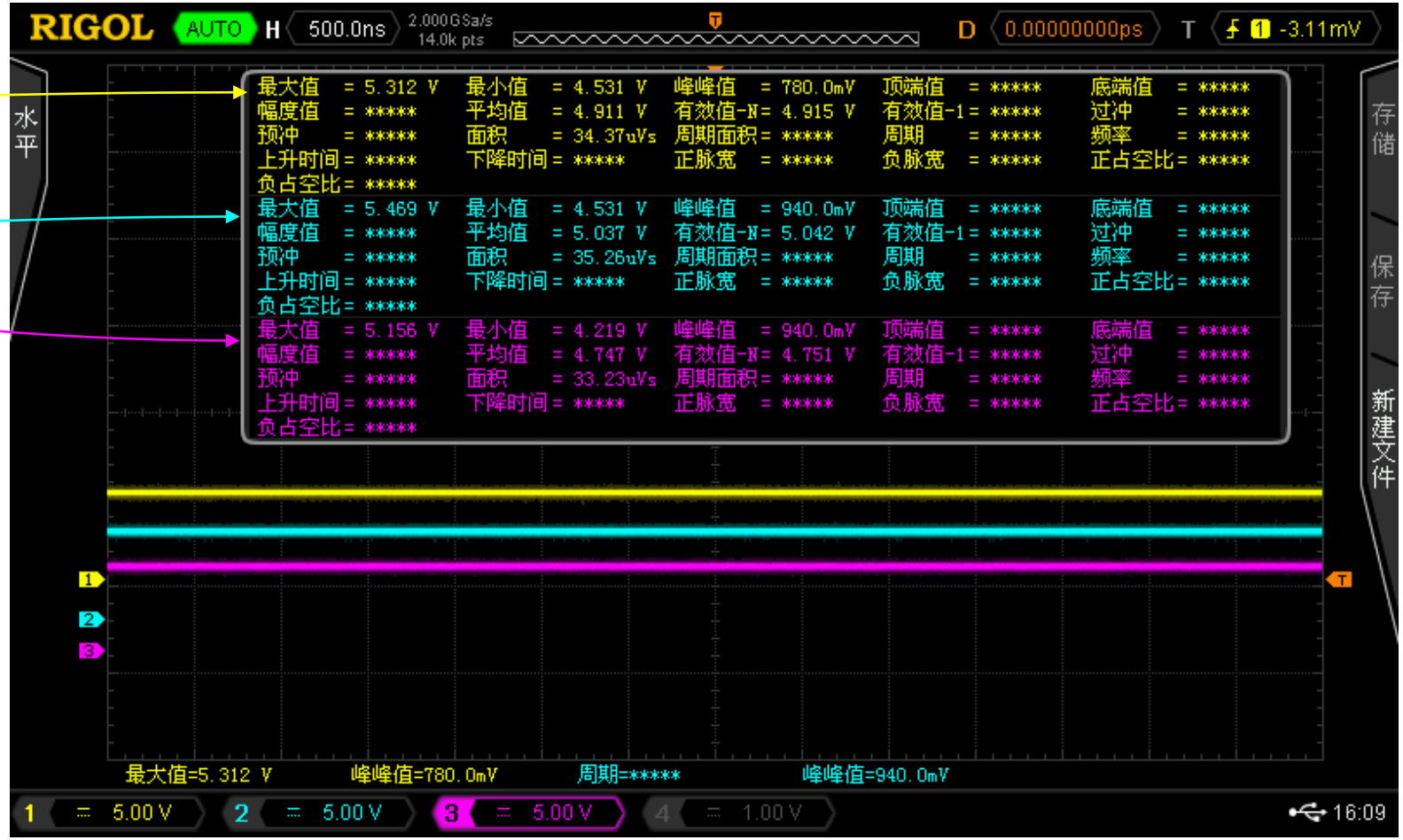
A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=0V
0V	5V	Vf=0V
5V	5V	Vf=5V

二极管设计之两输入与门电路--直流输入

■ 与门电路图、真值表

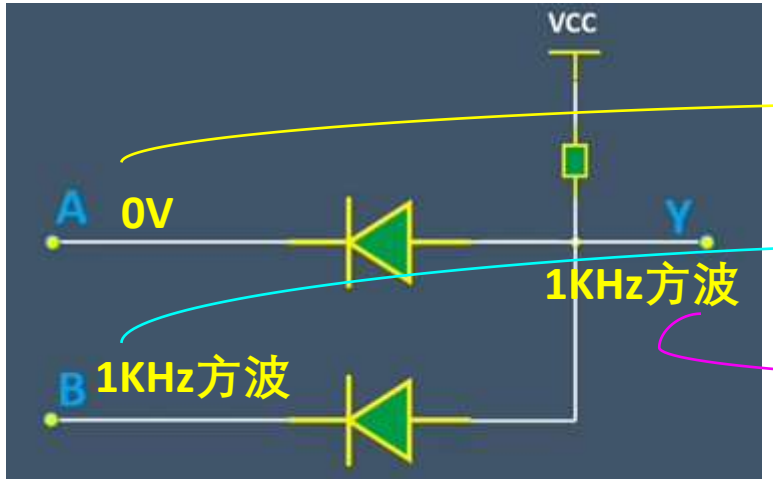


A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=0V
0V	5V	Vf=0V
5V	5V	Vf=5V

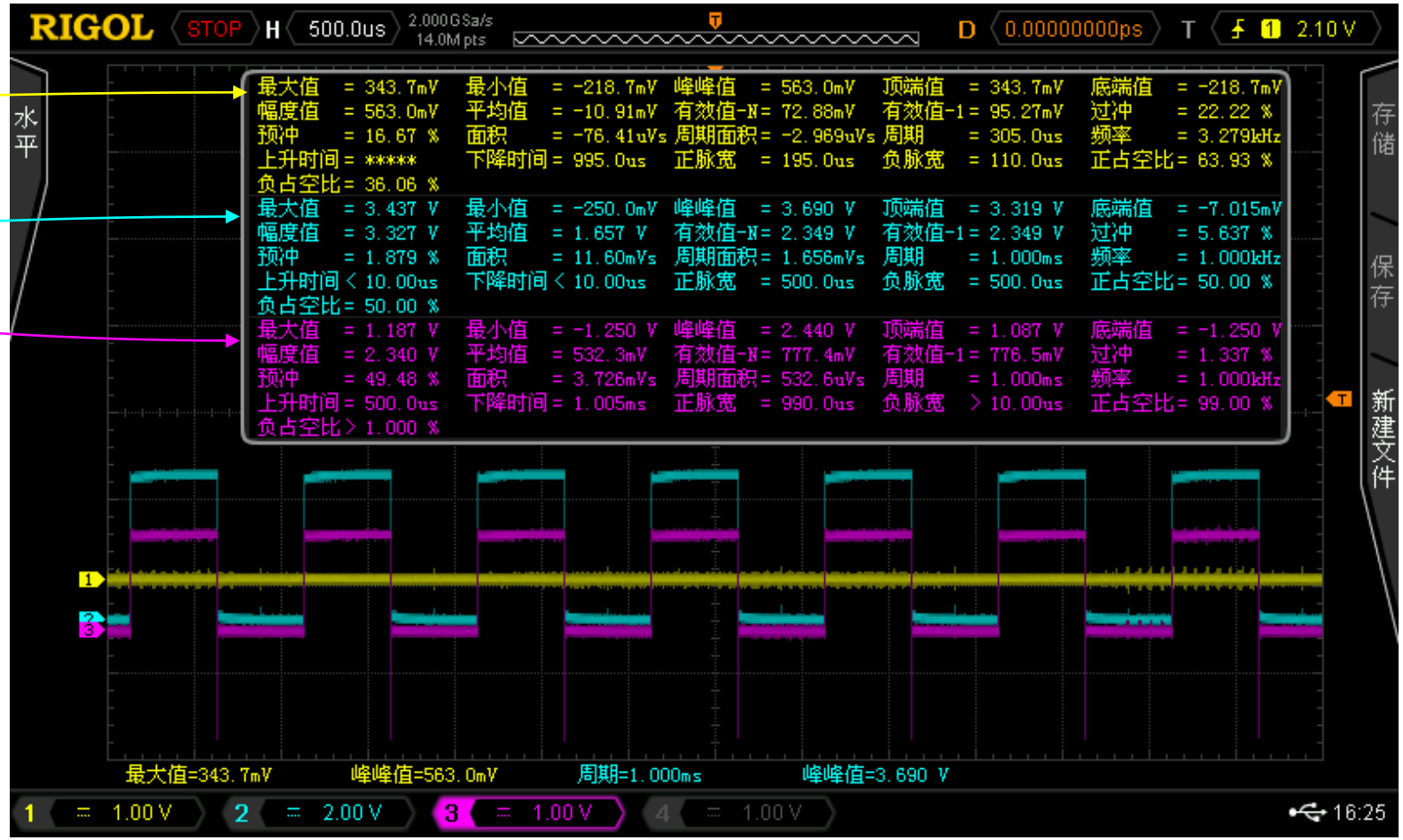


二极管设计之两输入与门电路—方波输入

■ 与门输入为方波（占空比50%）

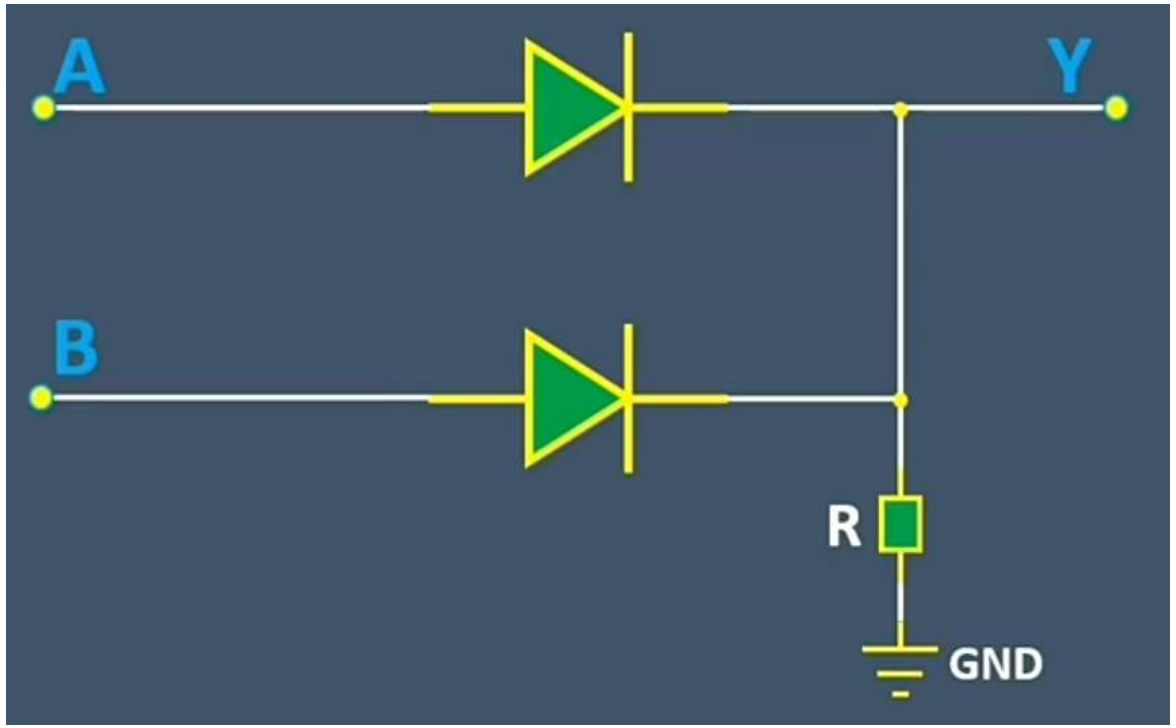


A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=0V
0V	5V	Vf=0V
5V	5V	Vf=5V



二极管设计之两输入或门电路--直流输入

■ 或门电路图、真值表

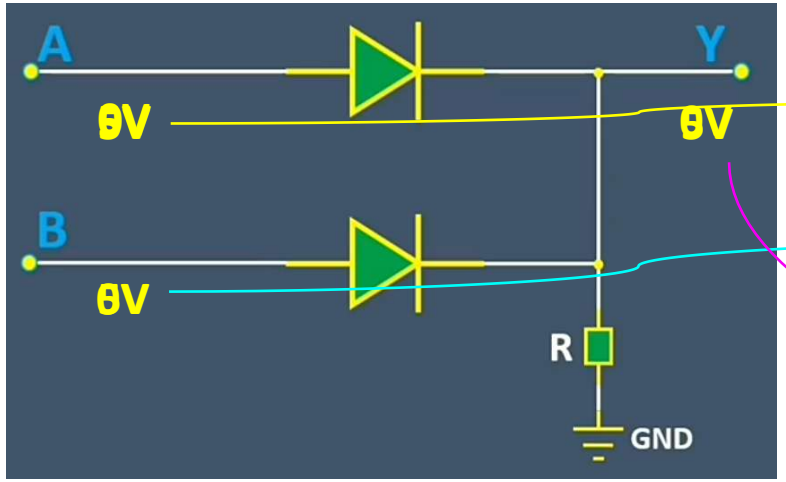


A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

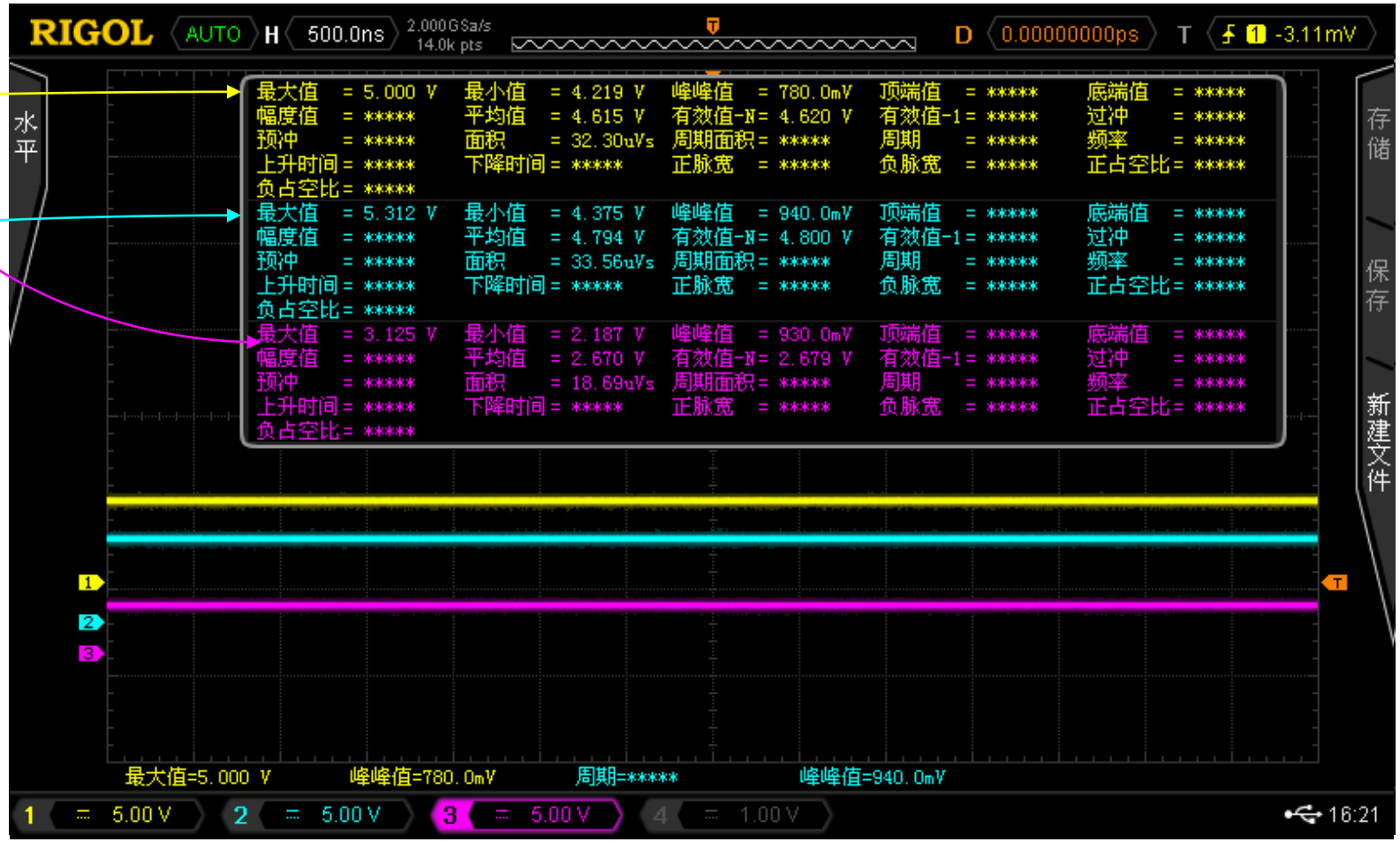
A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=5V
0V	5V	Vf=5V
5V	5V	Vf=5V

二极管设计之两输入或门电路--直流输入

■ 或门电路图、真值表

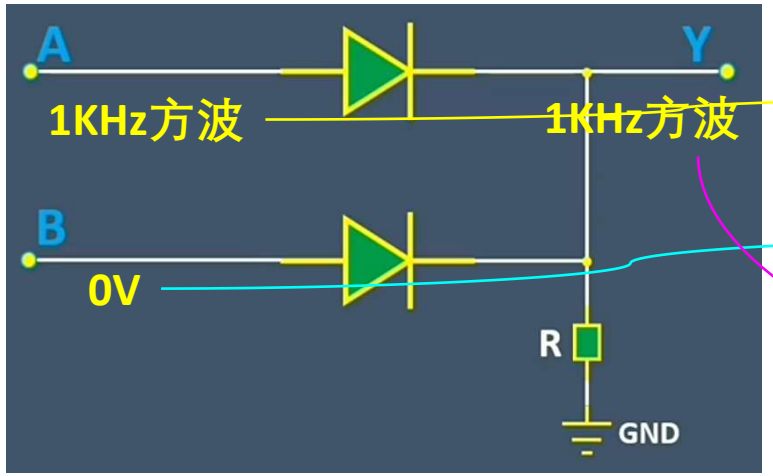


A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=5V
0V	5V	Vf=5V
5V	5V	Vf=5V

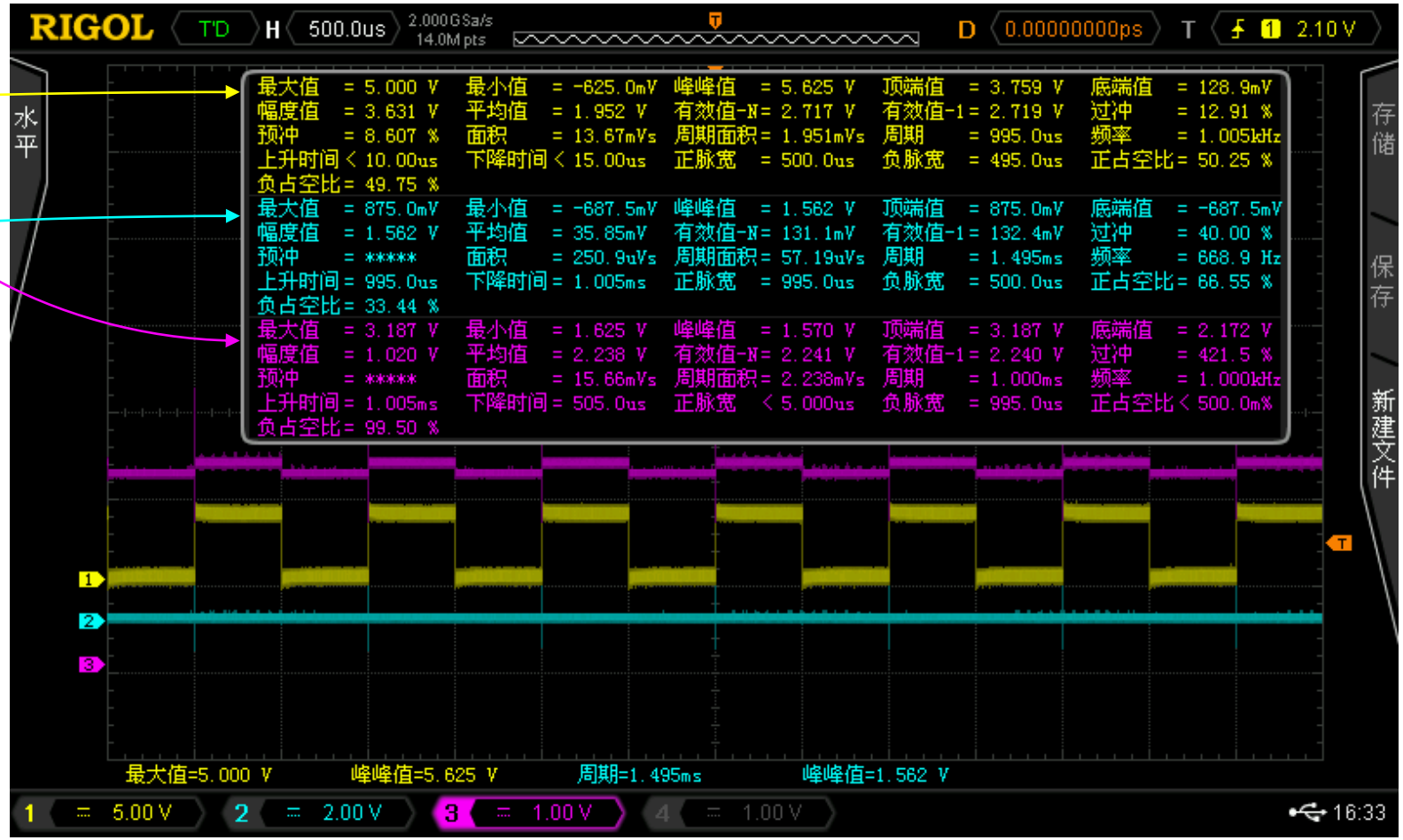


二极管设计之两输入或门电路—方波输入

■ 或门输入为方波（占空比50%）

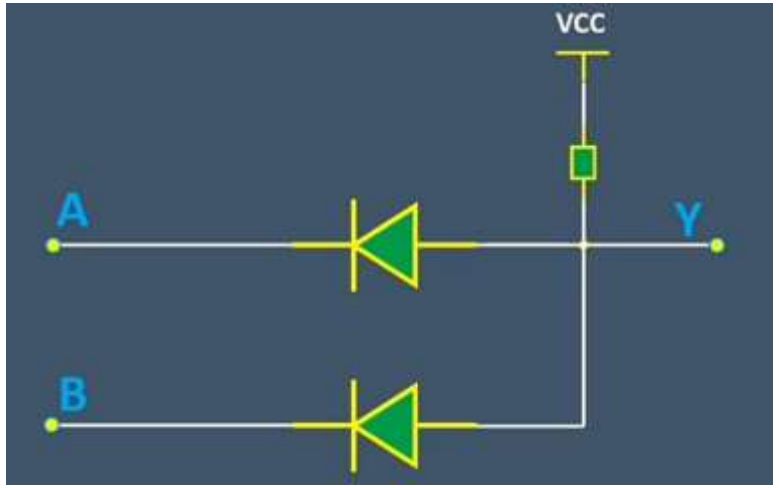


A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=5V
0V	5V	Vf=5V
5V	5V	Vf=5V



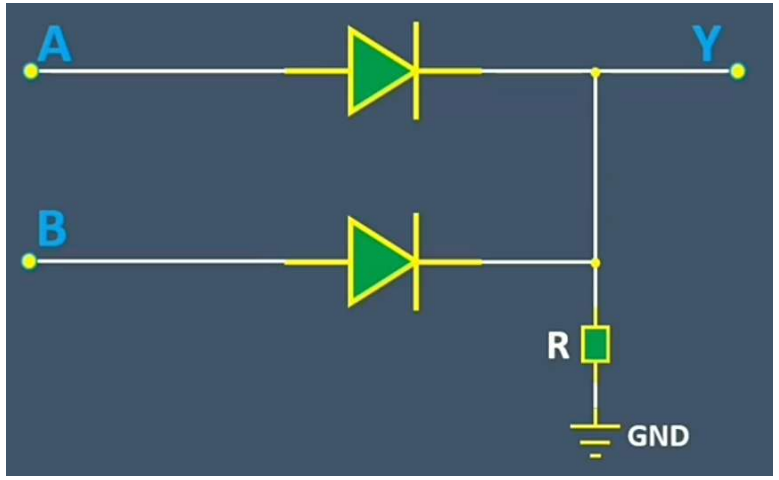
二极管设计之两输入与门、或门电路

■ 或门电路图、真值表比较



A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=0V
0V	5V	Vf=0V
5V	5V	Vf=5V



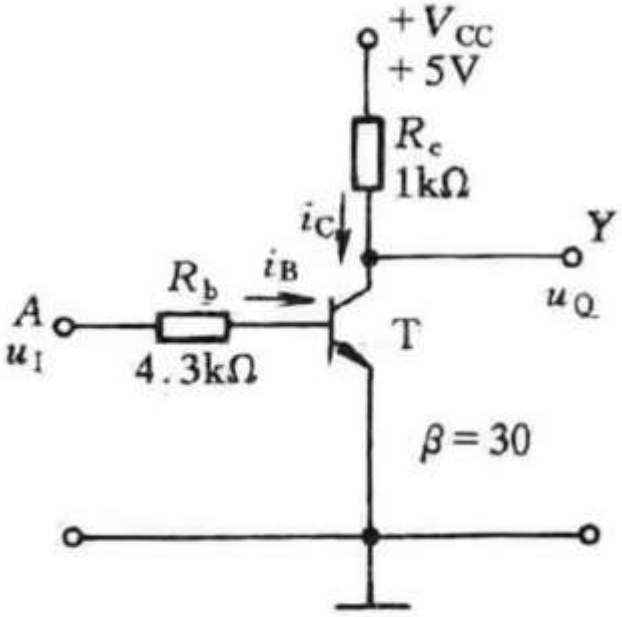
A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

A	B	Y
0V	0V	Vf=0V
5V	0V	Vf=5V
0V	5V	Vf=5V
5V	5V	Vf=5V

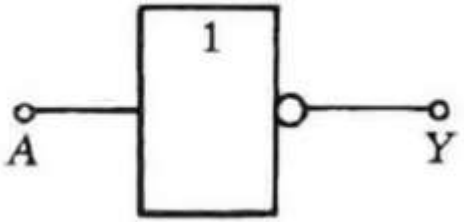
三极管设计之非门电路

■ 电路原理

- ➢ 当输入为高电平时，三极管导通，输出点电压为三极管的集电极和发射极之间的压降，即0.3V，即输出为低电平；
- ➢ 当输入为低电平时，三极管截止，输出电压为上拉的电压+5V，即输出为高电平



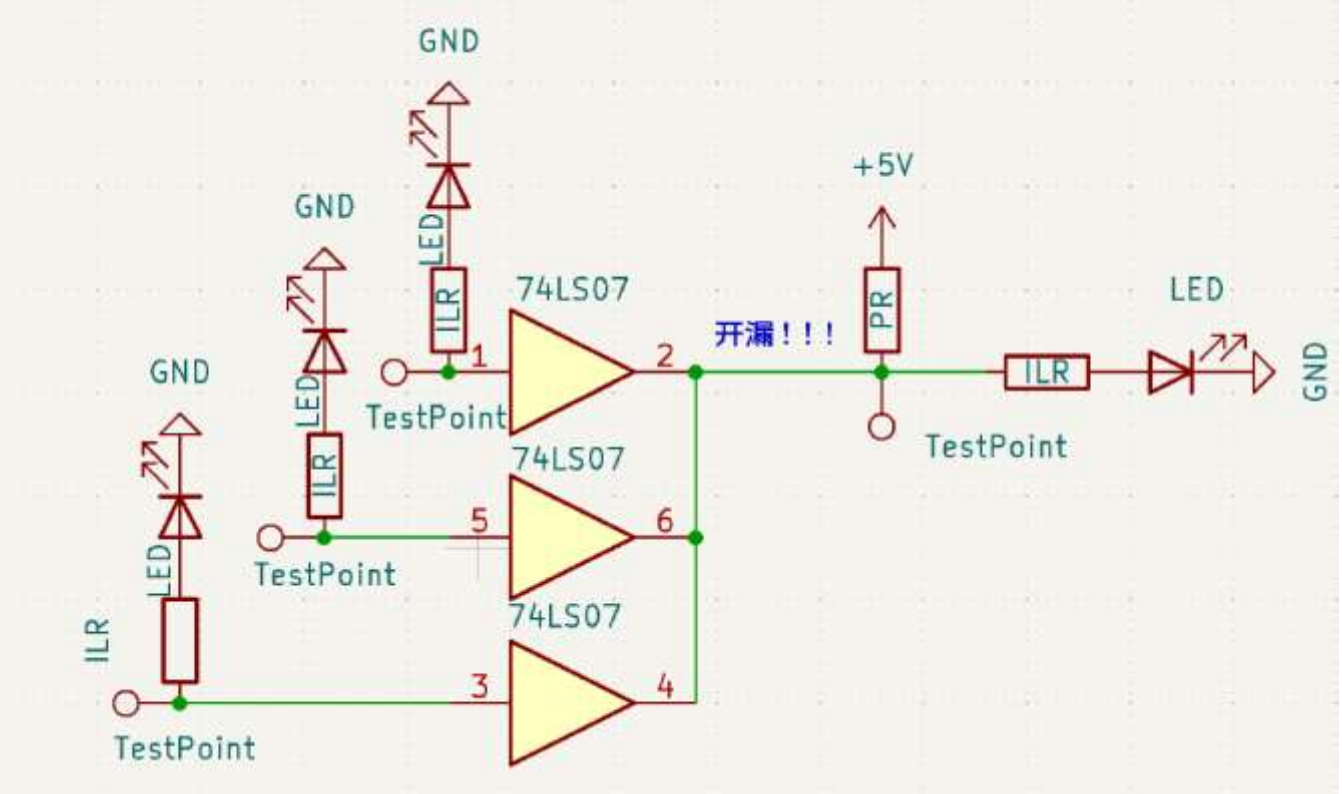
(a)



传输门线与电路

■ 电路原理

- 电路基于74LS07传输门，其输出端为开漏结构，意味着输出要么是低电平，要么是高阻态（可以理解为输出点与其他电路断开）
- 图中若任意一个输入为低则输出为低电平，LED两端无压降
- 图中若输入均为高，则输出端口为高阻态，相当于输出与其他电路断开，省略TestPoint左侧电路，LED被点亮



且听下节课继续分解！